

August 05, 2026
Ref.: HR/22013/26

To Whom It May Concern

Subject: Certificate of Internship

This is to certify that **Mr. Adam Ali Abdelrahman Zaatar** successfully completed his internship at ProgressSoft from **June 1, 2026, to August 6, 2026**. As part of the internship program, he has received practical training in the following areas:

Introduction to ProgressSoft:

- Introduction to ProgressSoft's main departments, business and technical teams
- Prepare development environment.

Technical Training:

Basics Developer setup

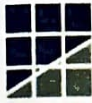
- Working with Posix environment.
- Prepare development environment
 - Installing Java SE 11+
 - Install IntelliJ
 - Install Maven
 - Install Git
 - Do the required SSH setup for Git
- Use Git as a source version control:
 - Create an account in a Git repository
 - Cloning projects from remote repositories
 - Understand how Git change sets and history work
 - Practice basic Git commands: add, commit, push, pull
 - Work with multiple branches and learn how to merge between them and fix conflicts

Java Programming Level 1

- Basic of Java SE 11+ programming syntax and OOP
 - Understand the difference between JDK and JRE
 - Understand how JVM works.
 - Data Types and scoping and access modifiers
 - The basics syntax (loops, conditions, Java Main application, Operators, arrays)
 - Working with String and StringBuilder
 - Understanding an Object life cycle
 - Annotations
 - Follow Java coding standard

Java Programming Level 2

- Generics
- Lambda expressions
- Implementing abstract classes and interfaces
- Overriding Object methods
- Polymorphism, overriding, and Overloading



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- Handling exceptions
- Inner classes
- Lambda Expressions
- Knowledge of Java APIs:
 - IO Streams
 - NIO Streams
 - Collections API
 - Streaming API
 - Concurrency API, utils, and Thread pools
 - String formatting, tokenizing, and pattern matching (Regex)
 - JDBC
 - Reflection API and Dynamic Proxy
- Maven as a build tool:
 - Understanding Maven build life cycle
 - Define project dependencies and understand the scope and how dependencies are cached in local repositories
 - Understand the difference between releases and snapshot releases
 - Connect with third party repositories using settings.xml
 - Define project plugins
 - Understand packaging types pom, war, and jar
 - Break project code into modules
 - Integrate Maven with Java IDEs
 - Create a template project using Maven Archetype plugin
- Follow Test-Drive Development (TDD):
 - Understand the concept of unit testing.
 - Learn how to implement a simple test case and manage to run it using junit-5 through Maven build life-cycle.
 - Follow TDD cycle: implement test case, make it fail, implement production code, make the test case success, then do required refactoring.
 - Understand how TDD should lead the OOP design.
 - Understand the concepts of Mocking and Spying.
 - Understand test class life cycle and use proper setup annotations.
 - Create parameterized and repeated tests
 - ProgressSoft Corporation 5
 - Use Mockito as a utility for mocking and spying
- Clean code principles:
 - Apply basic refactoring techniques
 - Extract methods, refactor long methods
 - Handling Duplicates
 - Introducing parameter objects
 - Extracting variables
 - Introducing classes
 - Introducing super classes
 - Inline members and classes
- Understand main clean code principles:
 - Encapsulate what varies
 - Favor composition over inheritance
 - Code to an interface



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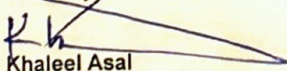
- Open for extension and closed for modification
- Understand SOLID principles:
 - Single responsibilities
 - Open for extension and close for modification
 - Liskov Substitution
 - Interface Segregation
 - Dependency Injection
- Understand Design patterns:
 - Strategy
 - State
 - Decorator
 - Factory
 - Adapter
 - Builder
 - Singleton

Work with JPA 2.1

- Define Entities and map Java types to tables and columns
- Understand the usage of EntityManager and transaction scopes
- Understand Entity lifecycle
- Define auto-generated keys
- Define embeddables
- Use JPQL and criteria API to query and update database
- Spring Framework
- Work Spring IoC container
 - Understand spring IoC container and its life-cycle
 - Understand bean definitions types and scopes
 - Use Annotations to configure a spring application and follow dependency injection best practices
- Work with Spring Boot:
 - Implement HTTP endpoints using Spring MVC
 - Implement a simple application to consume spring MVC endpoints
 - Use spring-data to create persistence layer
 - Use spring security to authenticate and authorize access to HTTP endpoints

During the period of his tenure at ProgressSoft, Mr. Adam was punctual, hardworking, and inquisitive. It has been a pleasure working with him. We wish him all the best in his future.

Sincerely,



Khaleel Asal

Human Capital Manager
ProgressSoft Corporation

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